

Notching, Boring and Crawl Space Clearances

What a tradesman is allowed to cut out of a structural member, and how far wood must stay from bare earth.

A joist carries load in bending, and in bending the top and bottom edges do almost all the work while the middle does very little. Every notching rule follows from that: keep the cut shallow, keep it short, and keep it out of the middle of the span where the bending is greatest.

The numbers here

Joist notch — maximum depth	1/6 of member depth
Joist notch — maximum length	1/3 of member depth
Joist notch — never located in	the middle 1/3 of the span
Bored hole — maximum diameter	1/3 of member depth
Bored hole — minimum distance from an edge, another hole or a notch	2 in
Non-bearing wall stud — maximum notch	40 %

YOU HAVE MISSED THIS

Notches in solid lumber joists, rafters and beams should not be located:
In the middle third of the span.

YOU HAVE MISSED THIS 2 TIMES

The maximum notch allowed in non-bearing wall studs is:
40%.

A bored hole may be larger than a notch is deep, which looks backwards until you picture it: a hole through the center removes material from the part that was doing least, while a notch cuts into the edge that was doing most. That is also why a hole must stay two inches clear of any edge or of another hole.

Wood against the ground

Wood in contact with damp earth decays, so every wood member in a crawl space has a minimum clearance. Learn it as a ladder, biggest to smallest.

The numbers here

Wood joists or the wood floor above exposed ground	18 in
Untreated girders above exposed earth	12 in
Wood framing resting on a concrete or masonry foundation wall	8 in
Air space at the end of a wood girder entering masonry	1/2 in

Eighteen, twelve, eight — joist, girder, framing. The half inch at the girder end is a different idea: it is an air gap so the buried end of the beam can dry rather than sit sealed in damp masonry.

Say it back

Cover the page. Answer each one out loud.

1. Give the three joist notch rules: depth, length, and location.
2. Why is a bored hole allowed to be bigger than a notch is deep?
3. Maximum bored hole diameter, and its minimum clearance from an edge?
4. Maximum notch in a non-bearing stud?
5. Say the crawl space clearance ladder: joist, girder, framing.
6. What is the half inch at the end of a girder for?